DAY 17 | PRE-READ

From My Laptop to the World

A light read before Day 17. About 15 minutes, no code required.

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Why read this?

So far, everything you built runs on your laptop or in Colab. Real users cannot reach it. This week you learn how teams ship software safely and often, and how the cloud gives you computers on demand.

1. DevOps: one team, one goal

Imagine a restaurant where the chefs cook, then hand plates to a separate team who decides weeks later whether to serve them. Food goes cold and nobody owns the result. DevOps joins the people who build software (development) with the people who run it (operations), so the same team cooks and serves.

2. Assembly lines for code: CI/CD

Car factories check every part automatically as it moves down the line. **Continuous Integration** does the same for code: every change is automatically built and tested. **Continuous Delivery** keeps the product always ready to ship at the press of a button.

> **Key idea:** small changes, checked automatically, shipped often, are far safer than big changes shipped rarely.

3. Locks checked on the line NEW (v4)

The same assembly line can also check for security problems: risky code, parts (libraries) with known flaws, and keys left lying around. Doing this early is called **DevSecOps**. It matters even more now that AI tools write some of the code.

4. Renting computers: the cloud

Option	Cloud name	You look after
Own the car	On-premises	Everything, including the building and hardware
Lease a car	IaaS (Infrastructure as a Service)	The operating system, software and data
Taxi	PaaS (Platform as a Service)	Just your app and its data
Bus	SaaS (Software as a Service)	Just your account and how you use it

5. Meet Azure and today's assessment

Microsoft Azure is one of the largest cloud providers. You organise everything into **resource groups**, like folders, inside a **subscription**, which is how usage is billed. Azure DevOps is Microsoft's tool set for planning work, storing code and running CI/CD pipelines. Today also includes Basics assessment 3 on testing, data engineering and DevOps.

6. Think about these before class

- What steps would you check by hand before releasing an app? Which could a machine check instead?
- Which apps that you use are SaaS?
- Why might a bank prefer IaaS, while a start-up prefers PaaS?

Kaizen (改善): continuous improvement